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EX NO: 4	OFDMA Resource Block Allocation
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AIM: To simulate and analyze OFDMA (Orthogonal Frequency Division Multiple Access) resource block allocation and evaluate system performance in terms of resource utilization and user throughput using MATLAB.

Procedure:

1. Define the system bandwidth and calculate the number of available resource blocks (RBs) based on RB size (e.g., 180 kHz per RB).
2. Identify the number of active users in the OFDMA system for resource sharing.
3. Allocate resource blocks to each user based on a predefined or equal distribution strategy.
4. Compute Resource Block Utilization (%) using the ratio of allocated RBs to total RBs.
5. Assign modulation efficiency values (e.g., QPSK, 16-QAM, 64-QAM) for each user.
6. Calculate User Throughput (Mbps) using allocated RBs and modulation efficiency.
7. Plot graphs for RB allocation and utilization/throughput analysis using MATLAB.
8. Analyze results to evaluate spectrum efficiency and system performance.

Objective 1 Matlab Code and Output:

```
clc; clear; close all;
```

```
% System parameters
```

```
bandwidth = 10e6; % 10 MHz
```

```
rb_size = 180e3; % 180 kHz per RB
```

```
% Total Resource Blocks
```

```
total_RB = floor(bandwidth / rb_size);
```

```
% Users
```

```

users = 1:6;

% RB allocation (simple equal + remainder distribution)
base = floor(total_RB / length(users));

alloc = base * ones(1,length(users));
alloc(1:mod(total_RB,length(users))) = alloc(1:mod(total_RB,length(users))) + 1;
utilization = (alloc / total_RB) * 100;

disp('Total Resource Blocks:')
disp(total_RB)

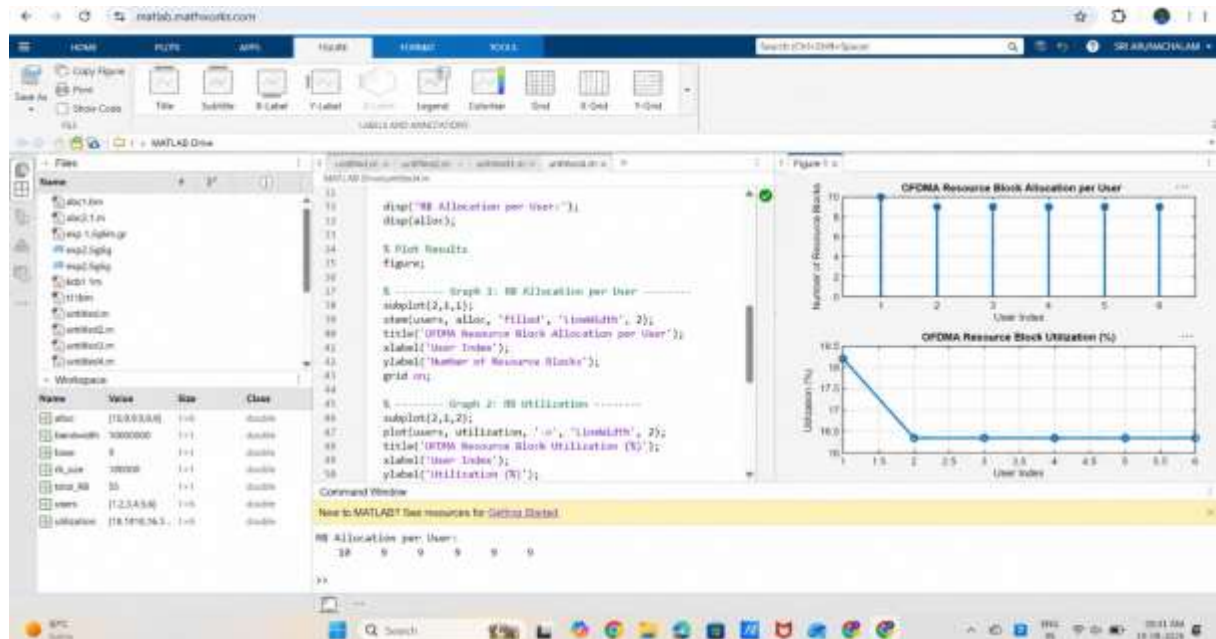
disp('RB Allocation per User:')
disp(alloc)

figure
% ----- Graph 1: RB Allocation per User -----
subplot(2,1,1)
stem(users, alloc, 'filled', 'LineWidth', 2)
title('OFDMA Resource Block Allocation per User')
xlabel('User Index')
ylabel('Number of Resource Blocks')
grid on

% ----- Graph 2: RB Utilization -----
subplot(2,1,2)
plot(users, utilization, '-o', 'LineWidth', 2)
title('OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid on

```

OUTPUT:



Result:

The first graph shows how available resource blocks are distributed among multiple users in the OFDMA system.